Shrikar Bende

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Summary

Final-year IT undergraduate at VIT Pune with hands-on experience in full-stack development, embedded systems, Artificial Intelligence, and Machine Learning. Strong analytical, problem-solving, teamwork, and adaptability skills, with a keen interest in building innovative technology solutions.

Education

Vishwakarma Institute of Technology, Pune, India **2023 – 2027 (Expected)**
- Bachelor of Technology (B.Tech), Information Technology | CGPA: **8.07/10**

Skills

Programming Languages: C++, SQL
Web Development: React.js, HTML5, CSS3, FastAPI
Databases & Tools: MySQL, Firebase
Cloud & DevOps: AWS, Docker, GitHub
AI & Agentic Frameworks: RAG Architecture, Transformer-based NLP, LLM APIs
Core Competencies: Data Structures and Algorithms (DSA), Object Oriented Programming (OOP), Database Management System (DBMS), Operating System (OS)

Projects

EchoVision – Intelligent Ultrasonic Glasses for Blind Navigation *(IEEE Xplore) (GitHub)*
- Designed a low-cost wearable assistive device using an ESP32 microcontroller and HC-SR04 ultrasonic sensor, achieving real-time obstacle detection with 2.25% MAPE and under 150 ms system latency.
- Integrated Firebase backend with GPS tracking, Web Speech API voice navigation, and a Leaflet.js map interface; 80% of visually impaired trial users found it easy to use after a 5-minute orientation.
- **Technologies:** ESP32, HC-SR04, Firebase, GPS, Leaflet.js, Web Speech API

AI-Based Bail Decision Intelligence System for Indian High Courts *(GitHub)*
- Built a full-stack legal AI platform that predicts bail outcomes and case duration using machine learning, with SHAP explanations of the top factors behind each prediction and a fairness audit across High Courts.
- Fine-tuned Legal-BERT for petition analysis and added vector-based precedent search, an LLM-powered legal research chat and automated court-document drafting, secured with JWT authentication.
- **Technologies:** Python, FastAPI, React, Legal-BERT, SHAP, Groq, Gemini, SQLite

Publications

Specialized LLMs for Hindi Medical NLP *(IEEE Xplore)* **2025**
- Implemented a multi-modal embedding layer combining clinical knowledge graphs, contextual embeddings, and Hindi medical tokenization for code-mixed clinical text.

EchoVision: Intelligent Ultrasonic Glasses for Enhanced Blind Navigation *(IEEE Xplore)* **2025**
- Co-authored IEEE-indexed paper designing a low-cost wearable assistive device using an ESP32 microcontroller and HC-SR04 ultrasonic sensor, achieving real-time obstacle detection with **2.25% MAPE** and under **150 ms** system latency.

Certifications

IBM Full Stack Software Developer (2025) *(Certificate)* | AWS Academy Graduate – Cloud Foundations (2025) *(Credential)* | Forage Virtual Experience Program (2025) *(Certificate)*

Extracurricular Activities

Social Welfare & Development – Pune Police Night Patrolling, VITSocials (2024): Night patrolling alongside Pune Police, building civic awareness and teamwork.